Wideline: Product Requirements

Revised Oct 19, 2026 · @Sasha · supersedes the "Scout" working draft of the same date

> **What changed in this revision.** Product named Wideline (subject to a trademark and domain check). The Wideline company is the responsible party under POPIA. Hosting moves from Azure to a small Hetzner server, with GPUs rented by the hour. Raw footage is kept for 7 days and clips for 14 days, unless favourited. Any registered scout can do review work. Departed players are deactivated, never deleted. The full list is in the *Decisions log* at the end. Rows marked CHANGED or NEW differ from the previous draft.

Overview

Wideline turns wide-angle match recordings from academies and amateur clubs into player stats and profiles that scouts can search. It starts free for invited pilot clubs, then grows into a paid scouting network.

Why the name. Wideline joins the two things the product is built on: the wide-angle camera every club already uses, and the touchline where scouts have always stood. It reads as football without being generic, and it works as a domain-style brand.

Vision. Every young player at a club that films its matches gets a verified, data-backed profile, so talent outside the professional system can be found.

Problem. Youth, academy, amateur and lower-league football produces almost no data. Scouts travel to watch a handful of matches and rely on word of mouth. Clubs already film matches, but nobody turns that footage into stats.

How it works. A club uploads a match recording and its lineups. A computer-vision (CV) pipeline detects players, the ball and key events. A registered scout picks the match up from the review queue and confirms or corrects anything the AI is unsure about. Stats then land on each player's profile, and scouts can search players whose families have opted in. CHANGED

Review work doubles as discovery. Scouts who review matches see players they might otherwise never hear about, which is the incentive for doing it.

Goals

1. Process a full 90-minute wide-angle match automatically, with a scout reviewing it in under 45 minutes during the pilot and under 30 minutes by v1.
2. Build verified player profiles for pilot academies and their squads.
3. Give scouts and recruiters a searchable pool of consented players, filtered by stats, position, age and region.
4. Be ready to switch on paid subscriptions without rebuilding anything.
5. Run the pilot on minimal fixed infrastructure cost, paying for GPU time only while a match is processing. NEW

Non-goals for now

- Broadcast TV footage, and phone footage filmed from the touchline.
- Native iOS and Android apps. Wideline is a responsive web app.
- Live, in-match analysis. Processing happens after the match.
- Replacing scouts' judgement. Wideline supports scouting; it does not rate potential or character.
- Professional league data, which can already be bought.
- A separate pipeline or rules for women's and girls' football. The same pipeline, stats and consent rules apply to every team. NEW
- Long-term storage of full match video. Raw footage is deleted after 7 days. NEW

Success metrics

Targets below are starting assumptions, to be agreed with the pilot clubs once they are chosen.

Metric	v0 Pilot target	v1 target
Pilot clubs actively uploading	5	25
Matches processed per week	10	75
Scout review time per match	under 45 min	under 30 min
Matches reviewed within 5 days of upload (before raw footage expires) NEW	90%	98%
Active reviewing scouts NEW	10	40
Goal and shot recall vs hand-tagged matches	95%	97%
Pass recall vs hand-tagged matches	not in scope	85%
Processing time per match (upload to review-ready)	under 4 h	under 2 h
Players consented to scout visibility	not in scope	60% of profiles
Weekly active scouts	10 (reviewers)	50

Users

Six personas, mapped onto the role set defined in AUTH-03. A single person can hold more than one role, for example a coach who also scouts.

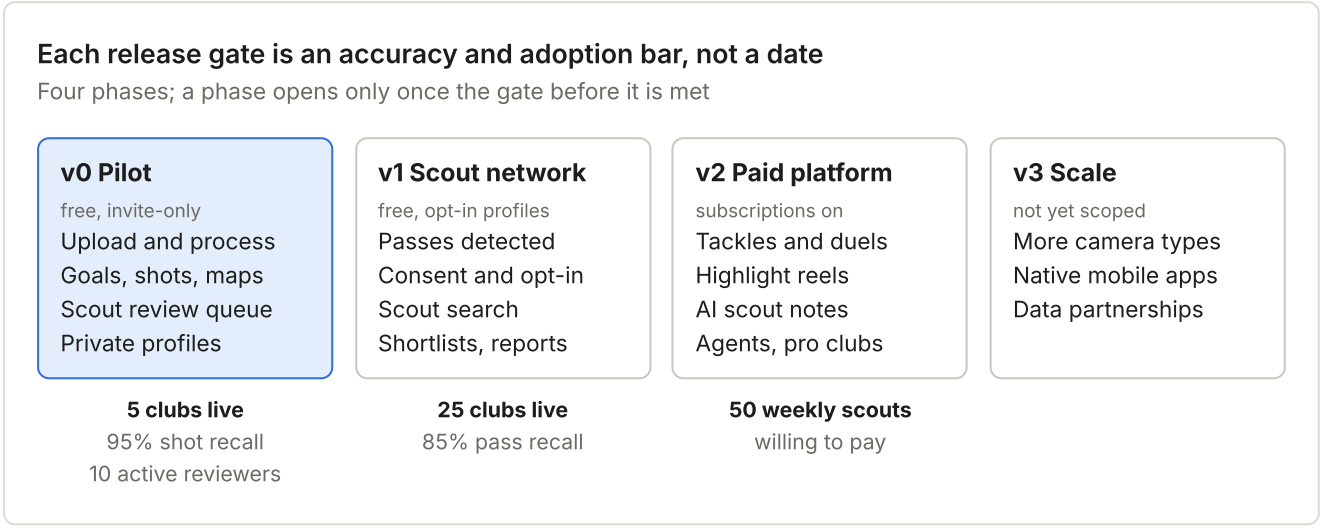
Role	Who they are	What they need	First release
Club admin	Runs an academy or amateur club	Set up the club, invite coaches, manage squads, hold consent records	v0
Coach	Coaches for a club	Upload matches and lineups, see how the squad is performing CHANGED	v0

Scout / recruiter CHANGED	Scouts for a club or works freelance	Review matches from the queue, then search consented players, build shortlists, write reports, watch clips	v0 (review), v1 (search)
Platform admin	Wideline staff	Onboard clubs and scouts, watch processing jobs, quality-check reviews, handle takedowns and support	v0
Player / parent	A player, or the guardian of one under 18	Control who sees the profile, view and share stats and clips	v1
Agent / pro-club recruiter	Represents players or recruits for a professional club	Discover players across many clubs, compare them, export shortlists	v2

What each one does most

- **Club admin.** Onboards the club, adds the squad, records guardian consent for under-18s, decides which coaches can upload, and marks players who leave the club.
- **Coach.** Uploads a recording after each match, enters the lineup, and checks the squad dashboard. Coaches no longer do review work.
- **Scout / recruiter.** The heaviest v0 user. Picks matches from the review queue, links tracks to players, confirms or corrects events, and favourites clips worth keeping. From v1, also searches, shortlists and writes reports.
- **Player / parent.** Accepts an invitation from the club, gives or withdraws consent, favourites their own clips, and shares a profile link with other clubs.
- **Agent / pro-club recruiter.** Works across the whole consented pool rather than one club, compares players, and exports a shortlist.
- **Platform admin.** Approves new clubs and scouts during the invite-only phase, spot-checks reviews, re-runs failed jobs, and acts on consent withdrawals and takedown requests.

Release roadmap



Wideline release roadmap · 4 phases, 3 gates

Each phase opens only when the gate before it is met, so the roadmap is driven by accuracy and adoption rather than by calendar dates. v0 stays invite-only until the CV pipeline is trusted; v2 only switches on billing once scouts have shown they will pay.

Because scouts now do the review work, scouts join in v0 rather than v1. In v0 they review matches and see the players in them, but cannot search the wider pool until v1.

Money follows the same rule. Everything is free through v0 and v1, which builds the pool of players that makes the product worth paying for. Billing and entitlements are built early but stay behind a feature flag until the v2 gate.

Functional requirements

Every requirement has an ID. The build backlog maps each chunk back to these IDs.

AUTH — accounts, organisations, roles

ID	Requirement	Release
AUTH-01	Sign up and sign in with email, magic link or password, plus Google	v0
AUTH-02	An organisation holds users, teams, players and matches; all data is scoped to it	v0
AUTH-03	Roles: owner, admin, coach, scout, agent, player, guardian, platform admin CHANGED (analyst merged into scout)	v0
AUTH-04	Invite users by email, with a role; invitations expire after 14 days	v0
AUTH-05	A user may belong to several organisations and switch between them	v0
AUTH-06	Every permission check runs server-side; no client-only gating	v0
AUTH-07	Audit log of logins, role changes, consent changes, exports and every review session	v0
AUTH-08	Scouts register independently, without belonging to a club, and are approved by a platform admin during the invite-only phase CHANGED	v0
AUTH-09	Scout registration verifies identity (name, ID or passport number, phone) and requires acceptance of the reviewer confidentiality terms before any review access NEW	v0
AUTH-10	A scout declares club affiliations and represented players at registration; these are used to block conflicted review assignments NEW	v0

SQUAD — teams, players, lineups

ID	Requirement	Release
SQUAD-01	Create teams within an organisation, with age group, gender and season. Gender is a label only; it does not change processing	v0
SQUAD-02	Add players with name, date of birth, position, preferred foot, height and squad number	v0
SQUAD-03	Import a squad from CSV, with a preview and per-row errors before committing	v0
SQUAD-04	Squad numbers are unique per team per season, and may change between matches	v0
SQUAD-05	Record a lineup per match: starters, substitutes, shirt numbers, positions	v0
SQUAD-06	Record substitutions with the minute, so minutes played are correct	v0
SQUAD-07	Opposition players may be added as unnamed placeholders	v0
SQUAD-08	Merge duplicate player records, keeping the full history	v1

SQUAD-09	A club admin can mark a player as departed. The player is deactivated from the club: removed from squads, lineups pickers and club dashboards, and no longer editable by the club. No data is deleted NEW	v0
SQUAD-10	A departed player keeps their full stat history and profile. With consent, a new club can claim the profile, so history carries over to the new club NEW	v1

MATCH — matches and footage

ID	Requirement	Release
MATCH-01	Create a match with date, venue, competition, home and away teams	v0
MATCH-02	Upload video in chunks, resumable, up to 8 GB, in MP4 or MOV	v0
MATCH-03	Validate on upload: duration, resolution at least 1080p, frame rate, aspect ratio	v0
MATCH-04	Reject or warn on footage that is not wide-angle, with guidance on how to re-film	v0
MATCH-05	Mark kickoff and half-time, so video time maps to match minutes	v0
MATCH-06	Transcode to HLS at several bitrates for playback in the browser	v0
MATCH-07	Show upload and processing status, with an estimated finish time and a countdown to raw-footage expiry CHANGED	v0
MATCH-08	Delete a match and everything derived from it	v0
MATCH-09	Import footage from a club's Veo or Pixellot account	later

CV — the computer-vision pipeline

ID	Requirement	Release
CV-01	Queue a processing job per match and run it on an hourly-rented GPU worker that starts for the job and shuts down after CHANGED	v0
CV-02	Detect players, goalkeepers, referees and the ball in every sampled frame	v0
CV-03	Track each player across frames, giving a track ID per player	v0
CV-04	Assign tracks to teams by kit colour, handling goalkeepers separately	v0
CV-05	Calibrate the pitch once per venue and camera, mapping pixels to pitch coordinates	v0
CV-06	Link tracks to lineup players, using shirt numbers where readable plus reviewer input	v0
CV-07	Detect goals and shots, each with a confidence score and a video timestamp	v0
CV-08	Produce per-player position tracks, heatmaps and distance covered	v0
CV-09	Write results as a versioned JSON artifact, so a re-run can be compared with the last	v0
CV-10	Retry a failed job up to 3 times, then raise it to a platform admin	v0

CV-11	Cut a short clip for every detected event at processing time, so clips survive deletion of the raw footage NEW	v0
CV-12	Detect passes with a sender, receiver and outcome	v1
CV-13	Detect tackles and duels, with the players involved and the outcome	v2
CV-14	Re-run the pipeline on a match with a newer model version. Only possible while raw footage still exists (7 days); older matches keep their stats as they are CHANGED	v2

REVIEW — human correction by scouts

ID	Requirement	Release
REVIEW-01	A shared review queue of processed matches. Scouts claim a match; a claim expires after 48 hours if unfinished, and the match returns to the queue CHANGED	v0
REVIEW-02	A scout cannot claim a match involving a club they are affiliated with, or a player they represent (AUTH-10) NEW	v0
REVIEW-03	Within a claimed match, list everything below the confidence threshold. Confirm, correct or reject each event, with the clip playing beside it	v0
REVIEW-04	Reassign an event to a different player	v0
REVIEW-05	Add an event the AI missed, by marking the video timestamp. Only possible while raw footage exists	v0
REVIEW-06	Fix a track-to-player link, which re-attributes every event on that track	v0
REVIEW-07	Keyboard shortcuts for confirm, reject and next, so review stays fast	v0
REVIEW-08	Mark a match reviewed, which publishes its stats to player profiles	v0
REVIEW-09	Every correction is stored as training data for later model improvement	v0
REVIEW-10	Auto-accept thresholds are set per organisation, per event type, in organisation settings	v0
REVIEW-11	Review playback is stream-only and watermarked with the reviewer's name; footage cannot be downloaded, and access ends when the review is submitted or the claim expires NEW	v0
REVIEW-12	The club can dispute a published review within 7 days; a platform admin decides. A sample of reviews is spot-checked against the footage each week NEW	v0
REVIEW-13	Matches nearing raw-footage expiry rise to the top of the queue, and platform admins are alerted 48 hours before an unreviewed match's footage is deleted NEW	v0
REVIEW-14	Show reviewer, time taken and correction count per match, plus a per-scout accuracy score from spot-checks	v1

PROFILE — player profiles and stats

ID	Requirement	Release
PROF-01	A profile per player: bio, club, position, and career totals	v0
PROF-02	Per-match stat lines, and totals and per-90 rates across a season	v0
PROF-03	Heatmap and average position per match and per season	v0
PROF-04	Every stat traces back to the events behind it, each with a match timestamp. Each event plays its clip while the clip is retained, and permanently if it is favourited CHANGED	v0
PROF-05	Profiles are private to the club until consent makes them visible	v0
PROF-06	Only matches marked reviewed count toward published stats	v0
PROF-07	Each stat shows how many matches it is based on, so small samples are obvious	v0
PROF-08	Favourite a clip to keep it beyond 14 days. Club staff, the reviewing scout, and the player or guardian can favourite. Favourited clips count against an entitlement limit NEW	v0
PROF-09	A departed player's profile shows their former club and history, and remains under the player's or guardian's control NEW	v0
PROF-10	A shareable public profile link the player or guardian can switch off	v1
PROF-11	Compare up to 4 players side by side	v2
PROF-12	Auto-generated highlight reel of a player's favourited clips CHANGED	v2

SEARCH — discovery

ID	Requirement	Release
SEARCH-01	Search consented players by name, position, age band, region and club (including "no current club")	v1
SEARCH-02	Filter by stat thresholds, such as goals per 90 above a value	v1
SEARCH-03	Results exclude any player without current consent	v1
SEARCH-04	Sort by any stat, with a minimum-matches filter	v1
SEARCH-05	Save a search and get alerts when new players match it	v2

SCOUT — shortlists and reports

ID	Requirement	Release
SCOUT-01	Create named shortlists and add or remove players. In v0, only consented players from matches the scout has reviewed CHANGED	v0
SCOUT-02	Write a scouting report against a player, with ratings and free text	v1
SCOUT-03	Reports are private to their author unless shared	v1

SCOUT-04	Share a shortlist with a colleague by link or invitation	v1
SCOUT-05	Export a shortlist or report to PDF or CSV	v2

CONSENT — privacy and POPIA

ID	Requirement	Release
CONS-01	Every player record stores one of five consent states: none, requested, club-only, scout-visible, withdrawn	v0
CONS-02	Players under 18 need consent from a parent or guardian, captured with name, relationship, timestamp and method	v0
CONS-03	Consent can be withdrawn at any time, taking effect on every live query immediately	v0
CONS-04	Consent wording states that matches are reviewed by independent registered scouts, and that data is stored and processed outside South Africa NEW	v0
CONS-05	Guardians get an emailed consent request with a plain-language explanation of what is collected	v1
CONS-06	Players turning 18 are prompted to confirm consent in their own right	v1
CONS-07	Export everything held about a player, on request	v1
CONS-08	Delete a player and all derived data, on request, within 30 days. This is separate from leaving a club, which never deletes data	v1
CONS-09	Retention policy: raw footage and HLS renditions are deleted 7 days after upload; event clips 14 days after upload unless favourited; stats are kept CHANGED	v0
CONS-10	Faces in shared clips can be blurred for non-consented players	v2

NOTIFY — notifications

ID	Requirement	Release
NOTIF-01	Email when processing finishes or fails	v0
NOTIF-02	Email scouts when new matches enter the review queue, and email the club when its match has been reviewed NEW	v0
NOTIF-03	Warn the club and player or guardian 3 days before un-favourited clips are deleted NEW	v0
NOTIF-04	In-app notification centre	v1
NOTIF-05	Email a guardian when consent is requested, granted or withdrawn	v1
NOTIF-06	Notify a player when a scout views their profile, if they have opted in	v2

ADMIN — platform operations

ID	Requirement	Release
ADMIN-01	Approve or reject new organisations and scouts during the invite-only phase	v0
ADMIN-02	Dashboard of processing jobs, with the ability to re-run or cancel	v0
ADMIN-03	Act on a takedown or consent-withdrawal request	v0
ADMIN-04	Feature flags per organisation	v0
ADMIN-05	Usage reporting per organisation: matches, minutes, GPU cost, storage	v0
ADMIN-06	Review-queue dashboard: unclaimed matches, time to expiry, reviewer activity, disputed reviews; suspend a scout's review access NEW	v0

BILL — subscriptions, built early, switched on later

ID	Requirement	Release
BILL-01	Plan and entitlement model: every limit is read from entitlements, never hard-coded	v0
BILL-02	All pilot organisations run on a free plan with generous limits	v0
BILL-03	Payment provider integration behind a feature flag, switched off	v2
BILL-04	Self-serve checkout, plan changes and cancellation	v2
BILL-05	Enforce entitlement limits such as matches per month, scout seats and favourited clips	v2
BILL-06	Invoices and payment history	v2

Stats catalogue

Each stat has one definition, used by the CV pipeline, the review screen and the profile alike. Auto-accept is the confidence at or above which an event is published without review; anything below goes to the review queue.

Stat	Definition	How it is detected	Auto-accept	Release
Minutes played	Minutes on the pitch, from the lineup and substitutions	Entered by the club, not detected	n/a	v0
Distance covered	Metres run, from the player's position track	Position track plus pitch calibration	n/a	v0
Heatmap	Share of time spent in each zone of the pitch	Position track binned into a grid	n/a	v0
Average position	Mean pitch position while on the pitch	Position track	n/a	v0
Goal	The ball fully crosses the goal line and the restart follows	Ball track crossing the goal plane, plus a restart check	0.90	v0
Shot	A deliberate attempt on goal	Ball leaves a player toward goal above a speed threshold	0.75	v0
Shot on target	A shot that would enter the goal but for a save or a block	Ball trajectory against the goal frame	0.75	v0
Touch	A player makes contact with the ball	Ball track changing direction near a player	0.80	v0
Pass attempted	A deliberate ball release toward a teammate	Possession change between two players of one team	0.70	v1
Pass completed	A pass received by the intended teammate	As above, with the receiver on the same team	0.70	v1
Pass direction and length	Forward, sideways or backward; short, medium or long	Pitch coordinates of start and end	n/a	v1
Cross	A pass from a wide area into the box	Pass with start and end zones in defined areas	0.70	v1
Assist	The pass directly before a goal	Last completed pass before a goal, same team	0.80	v1
Possession share	Share of match time each team holds the ball	Aggregated from touches	n/a	v1
Tackle	A challenge that takes the ball from an opponent	Two opposing players close to the ball, then possession changes	0.65	v2

Duel won	Any contested ball won, in the air or on the ground	Proximity, contest, then possession outcome	0.65	v2
Interception	A pass cut out by an opponent	Pass whose receiver is from the other team	0.70	v2
Recovery	Winning a loose ball	Possession gained with no opponent contest	0.65	v2

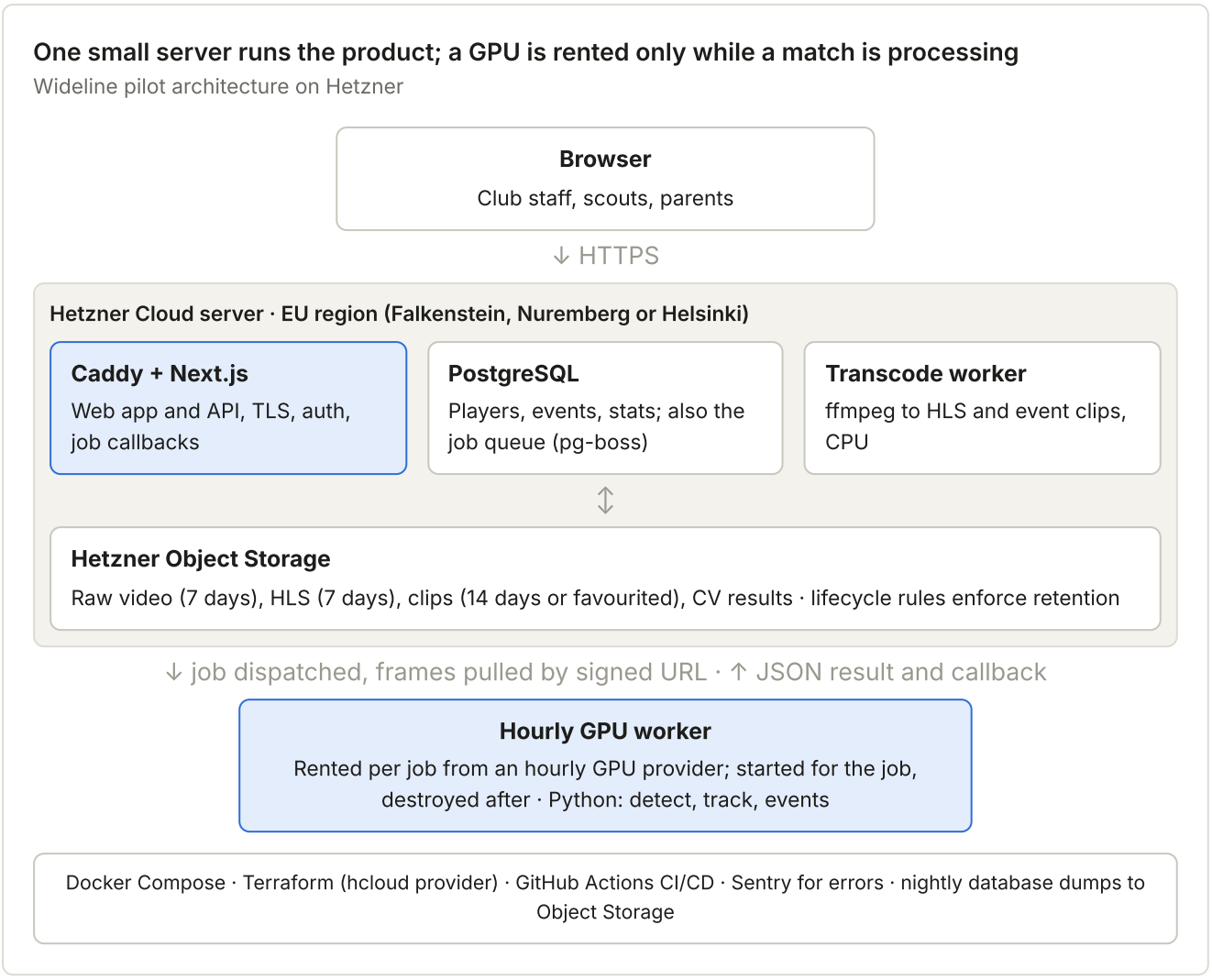
Rules that apply to every stat

- **Nothing is published unverified.** A stat only reaches a profile once its match is marked reviewed (REVIEW-08).
- **Every stat is traceable.** Each one links to its events, and each event to a match timestamp. A clip is shown while it is retained or if it was favourited (PROF-04).
- **Per-90 rates need a minimum sample.** Below 3 matches, the profile still shows the rate but flags the small sample beside it (PROF-07).
- **Opposition stats are collected but not attributed.** Without a named opposition lineup, events are recorded against placeholder players.
- **Thresholds are configurable per organisation** (REVIEW-10). A club that wants tighter data can raise them; a club that wants less review can lower them.
- **One pipeline for every team.** Men's, women's, boys' and girls' football use the same models, stats and thresholds.

Deliberately excluded

Expected goals (xG), progressive carries and pressures are left out of v0 to v2. Each needs either a large training set or reliable event chains that wide-angle footage at this level does not yet support. Shipping them inaccurately would cost more trust than they add.

System architecture



Wideline architecture · request path and job path

The web app never does heavy work. It writes a job to the queue and returns. For a CV job, a dispatcher starts a GPU instance by API; the worker pulls frames from Object Storage by signed URL, writes its JSON result back, calls the API, and the instance is destroyed. The expensive part stays isolated and is paid for by the minute rather than by the month.

This is a deliberately minimal starting point. It is one server, so it is a single point of failure, and it is sized for the pilot, not for v1 volumes. Everything runs in containers and is described in Terraform, so moving to a larger Hetzner server, splitting the database out, or moving to another cloud later is a redeploy, not a rewrite.

Stack

Layer	Choice	Why
Front end and API	Next.js (TypeScript), App Router	One codebase for pages and API routes; server components keep player data off the client

Hosting CHANGED	One Hetzner Cloud server at the cheapest tier that fits (shared vCPU, 4 GB RAM to start), Docker Compose	Lowest fixed cost while the product finds its feet. Resize the server in place when load requires it
Database CHANGED	PostgreSQL in a container on the same server, with a Hetzner Volume for data	Relational data with JSONB for raw CV output; Prisma for the schema and migrations. Move to a dedicated database server before v1
Object storage CHANGED	Hetzner Object Storage (S3-compatible)	Raw video, HLS renditions, clips and CV results. Lifecycle rules enforce the 7- and 14-day retention
Queue CHANGED	pg-boss on PostgreSQL	Durable jobs with retries and a dead-letter state, without running another service
CV workers CHANGED	Python container on an hourly-rented GPU, started per job by API	Hetzner's entry cloud servers have no GPU and its GPU servers are billed monthly, so the pilot rents by the hour. Provider chosen on price, API control and region (open question)
Video delivery CHANGED	HLS served from Object Storage through short-lived signed URLs	No CDN needed at pilot volumes; add one in front of Object Storage when playback demand grows
Sign-in CHANGED	Auth.js (or an equivalent open-source library) inside the Next.js app	No paid identity service; supports email magic links, passwords and Google
Infrastructure CHANGED	Terraform with the hcloud provider, plus Docker Compose	Every environment described in code
Observability CHANGED	Structured JSON logs, Sentry for errors, simple uptime monitoring	Enough to reconstruct a job's history without a paid monitoring stack

Data residency and cross-border transfer

Hetzner has no data centre in South Africa. Its locations are in Germany and Finland, plus cloud regions in the US and Singapore. Wideline will host in an EU region, so **all personal information, not only video frames, is stored outside South Africa**. The hourly GPU worker may also run outside the EU, depending on the provider chosen.

Under POPIA, transfers of personal information out of South Africa are restricted and need a lawful basis, such as the data subject's consent or binding agreements with the recipients. Wideline relies on both. The consent wording discloses offshore storage and processing (CONS-04). Operator agreements are signed with Hetzner and the GPU provider. Hosting in the EU also means data sits under GDPR-level protection. This position must be confirmed by a South African privacy lawyer.

Moving data back to South Africa later remains possible, because nothing in the build depends on a Hetzner-specific service except S3-compatible storage.

Data model

PostgreSQL, with Prisma owning the schema and migrations. Every table except User carries an `organisationId` and is filtered by it on every query, except review data, which is scoped to the claiming scout.

Entity	Key fields	Notes
Organisation	name, type (academy, club, agency), country, region, status, planId	The tenant boundary
User	email, name, authProviderId	Global; belongs to organisations through Membership
Membership	userId, organisationId, role, invitedBy, acceptedAt	A user may hold several
ScoutProfile NEW	userId, verifiedAt, idNumberHash, phone, termsAcceptedAt, status, accuracyScore	Independent scouts; status: pending, approved, suspended
ScoutAffiliation NEW	scoutUserId, organisationId or playerId, type, declaredAt	Blocks conflicted review claims
Team	organisationId, name, ageGroup, gender, season	gender is a label; no behaviour depends on it
Player CHANGED	organisationId, teamId, firstName, lastName, dateOfBirth, position, preferredFoot, heightCm, squadNumber, consentState, clubStatus, departedAt	dateOfBirth drives the under-18 consent rule. clubStatus: active, departed. Departing never deletes data
Consent	playerId, state, grantedBy, relationship, method, wordingVersion, grantedAt, withdrawnAt, evidenceUrl	Append-only; the current state is the newest row
Match	organisationId, homeTeamId, awayTeamId, playedAt, venue, competition, status	status: draft, uploading, processing, awaiting_review, in_review, published
Lineup	matchId, playerId, shirtNumber, position, startedAt, endedAt, isStarter	startedAt and endedAt give minutes played
Video CHANGED	matchId, blobPath, durationSec, width, height, fps, hlsManifestPath, four half-time offsets, rawExpiresAt, deletedAt	One per match. Record survives deletion of the file, so metadata stays
PitchCalibration	videoId, homographyMatrix, confidence, calibratedBy	Reusable per venue and camera
ProcessingJob	matchId, type, status, modelVersion, gpuProvider, startedAt, finishedAt, attempts, errorMessage, costCents	type: transcode, clips, cv

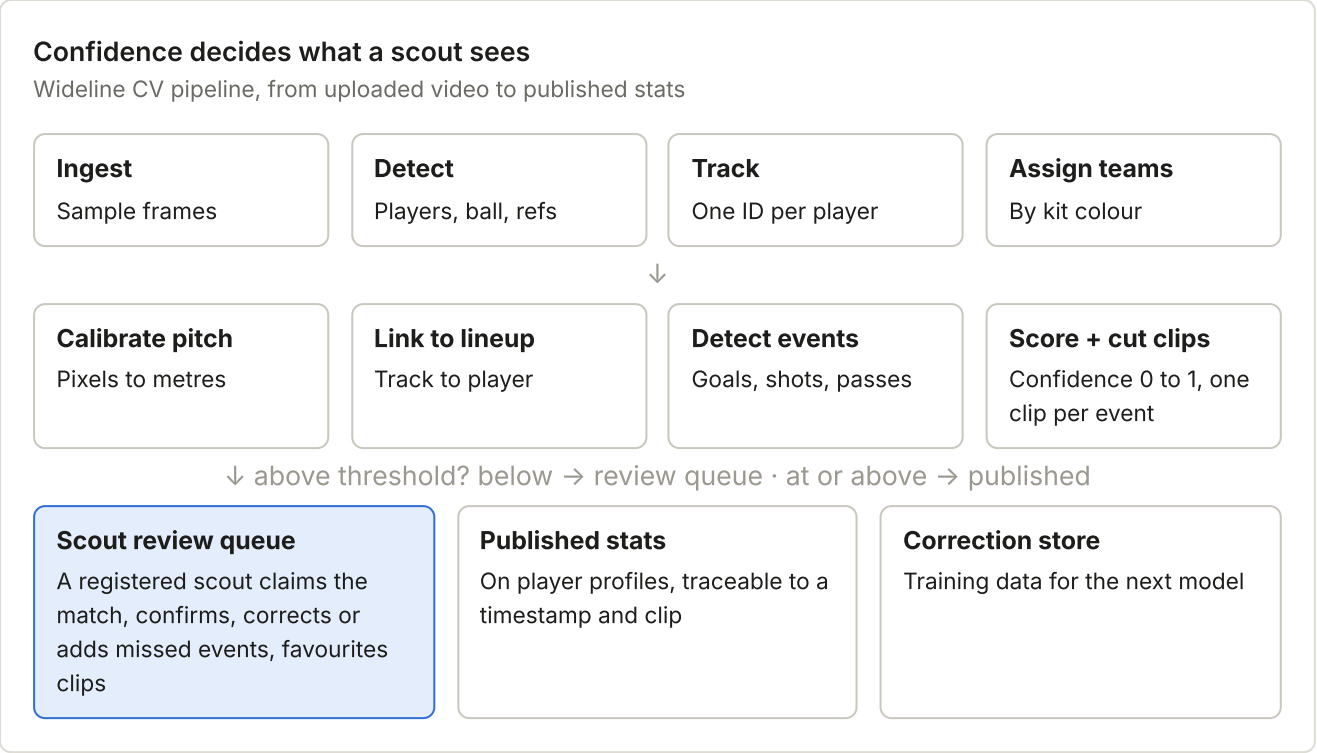
ReviewClaim NEW	matchId, scoutUserId, claimedAt, expiresAt, submittedAt, status, eventsReviewed, corrections	One active claim per match
ReviewDispute NEW	matchId, raisedBy, reason, resolvedBy, outcome	
Track	matchId, jobId, trackIndex, teamSide, playerId, linkConfidence, linkedBy	playerId null until linked
TrackPoint	trackId, timestampMs, pitchX, pitchY	High volume; partitioned by match
Event	matchId, type, playerId, trackId, timestampMs, pitchX, pitchY, confidence, status, payload	status: auto_accepted, pending_review, confirmed, rejected, manual
Clip NEW	eventId, blobPath, startMs, endMs, expiresAt, deletedAt	Cut at processing time
ClipFavourite NEW	clipId, userId, createdAt	Any favourite clears expiresAt; removing the last one restores it
EventCorrection	eventId, correctedBy, previousValue, newValue, correctedAt	Feeds the training set
PlayerMatchStats	playerId, matchId, minutesPlayed, goals, shots, passesCompleted, distanceM, heatmapBlobPath, stats (JSONB)	Derived; rebuilt when a match is re-reviewed. Every other stat lives in the JSONB map keyed by stat ID
Shortlist	ownerUserId, name, sharedWith	
ShortlistEntry	shortlistId, playerId, note, addedAt	
ScoutReport	authorUserId, playerId, matchId, ratings, body, visibility	
Plan	code, name, priceCents, currency, limits	limits is JSON
Subscription	organisationId, planId, status, provider, providerRef, currentPeriodEnd	
Entitlement	organisationId, key, value, source	Resolved from the plan plus any overrides
AuditLog	organisationId, actorUserId, action, targetType, targetId, metadata, createdAt	

Rules the schema must enforce

- A player's consent state is derived from the newest Consent row, never edited in place, so the history survives an audit.
- `Event.playerId` is only set once its Track is linked. Correcting a track link re-attributes every event on it in one transaction.
- `PlayerMatchStats` is a derived table, rebuilt from events whenever a match is published or re-reviewed. It is never written directly.

- `TrackPoint` is by far the largest table, roughly 22 players × 5 samples per second × 90 minutes, about 600,000 rows per match. Partition it by match. On a small server this is the first thing to watch for disk and memory.
- Marking a player departed changes `clubStatus` only. Nothing cascades.
- Deleting a player on request (CONS-08) cascades to lineups, tracks, events, clips and stats, and leaves a tombstone in `AuditLog`.
- Deleting a raw video or clip file sets `deletedAt` on its record. The record and every event and stat that came from it are kept.

CV pipeline



CV pipeline · 9 stages, 1 confidence gate

The reviewer never watches the whole match again. They see only the events the model was unsure about, each with its clip cued up. Every correction is kept, so the review burden should fall as the models improve.

The 7-day window. Because raw footage is deleted after 7 days, the full loop of upload, processing and review has to finish inside that window. Adding a missed event or re-running the pipeline needs the raw video. After day 7, a match's stats can still be corrected event by event, but nothing new can be found in it.

Models and licences

Licensing is a real constraint, not a detail. Ultralytics YOLO is AGPL-3.0: using it in a commercial SaaS means either open-sourcing the whole product or buying an Enterprise licence, and Ultralytics states that this applies even to custom-trained models and internal-only use. Build on Apache-2.0 models instead unless a licence is bought deliberately.

SoccerNet data and pretrained weights are mostly released for research. Treat them as a way to prototype and to benchmark, not as something to ship, until the terms are checked for each component.

Stage	Suggested starting point	Licence
Detection	RF-DETR (Roboflow) or RT-DETR	Apache-2.0, free commercially
Tracking	ByteTrack or BoT-SORT	Check per repo before adopting

Pitch calibration	Keypoint model from the SoccerNet calibration task	Verify dataset terms, which are often research-only
Shirt numbers	SoccerNet jersey-number recognition	Verify dataset terms
Event detection	Trained in-house on reviewed matches	Owned outright

The contract between the worker and the app

The CV worker writes one versioned JSON file to Object Storage and then calls the API. Nothing else passes between them, so the worker can be rewritten, or moved to another GPU provider, without touching the web app.

```
{
  "schemaVersion": "1.0",
  "jobId": "job_01HQ...",
  "matchId": "match_01HQ...",
  "modelVersions": { "detector": "rfdetr-1.2", "tracker": "bytetrack-0.4", "events": "wideline-events-
0.3" },
  "video": { "durationSec": 5820, "fps": 25, "sampledFps": 5 },
  "calibration": { "confidence": 0.91, "homography": [[1.0, 0.0, 0.0], [0.0, 1.0, 0.0], [0.0, 0.0, 1.0]]
},
  "tracks": [
    { "trackIndex": 14, "teamSide": "home", "shirtNumber": 7, "shirtConfidence": 0.42,
      "points": [{ "tMs": 12000, "x": 48.2, "y": 31.7 }] }
  ],
  "events": [
    { "type": "shot", "trackIndex": 14, "tMs": 1832400, "x": 88.1, "y": 34.0,
      "confidence": 0.81, "onTarget": true, "clipStartMs": 1827400, "clipEndMs": 1835400 }
  ],
  "warnings": ["ball not visible for 142s total"]
}
```

Pitch coordinates are in metres on a normalised 105 × 68 pitch, with the origin at the bottom-left corner of the attacking-left team. Every event carries a clip window; the CPU worker cuts the clips from it straight after the CV job (CV-11).

Confidence and review load

Thresholds from the stats catalogue decide what a scout sees. Set them high at first. A high auto-accept threshold means fewer events clear it, so scouts review more and the correction store fills quickly. Lower them per event type only once measured recall justifies it. Track reviewed items per match as a product metric: if it is not falling over time, the models are not learning from the corrections.

Non-functional requirements

Performance and cost

ID	Requirement
NFR-01	A 90-minute match is review-ready within 4 hours of upload in v0, and 2 hours by v1, including GPU start-up time
NFR-02	Page loads in under 2 seconds on a 4G connection from South Africa to the EU-hosted server
NFR-03	Video playback starts in under 3 seconds, seeking in under 1 second
NFR-04	GPU cost per match is measured and recorded per job, and reported per organisation
NFR-05	Target cost per match under R120 at pilot volumes, including GPU rental and storage for the retention period. Fixed monthly infrastructure (server, storage, backups) kept under R500 during the pilot CHANGED
NFR-06	Processing is idempotent, so a retried job cannot double-count events
NFR-07	A GPU instance that has been idle for 10 minutes, or has outlived its job by 30 minutes, is destroyed automatically, so a stuck job cannot run up a bill NEW

POPIA and privacy

This is the single biggest compliance risk, because most players will be children. Under POPIA section 34 a responsible party may not process a child's personal information at all, except under section 35. The relevant exception for Wideline is the first one: prior consent from a competent person, meaning the parent or guardian.

The Wideline company is the responsible party. It decides why and how player data is processed, so it carries the POPIA obligations directly. Clubs collect lineups and consent on Wideline's behalf. Hetzner, the GPU provider and any email provider are operators, each under a written operator agreement.

Section 34 bars processing, not publication, so a gate that merely hides an unconsented child's stats at publish time does not satisfy it: the derivation already happened. The pipeline therefore discards per-player data for non-consented minors at ingest. Whole-pitch detection and tracking still run, because they cannot be done selectively, but nothing is retained against a child without consent.

> **Scouts as reviewers is the new exposure.** To review a match, a scout sees footage of every child on the pitch, including children whose guardians have not consented to scout visibility. Wideline mitigates this in four ways. Scouts are verified, and they agree to binding confidentiality terms (AUTH-09). Playback is stream-only and watermarked (REVIEW-11). Access ends with the review, and every session is logged. The consent wording discloses that independent scouts review matches (CONS-04). Whether this is enough for footage of non-consented minors is the main question for the privacy lawyer.

ID	Requirement
NFR-08	No player under 18 has per-player data derived or retained without a recorded guardian consent. Detection and tracking run over the whole pitch, so the gate sits at ingest: tracks belonging to a non-consented minor are discarded before any event is attributed, not merely hidden at publication
NFR-09	Consent records store who granted it, their relationship to the player, the wording version shown, the method and the timestamp
NFR-10	Consent is specific: separate choices for club-internal stats and for scout visibility
NFR-11	Withdrawal is as easy as granting. It takes effect immediately on every live query, and within 24 hours on derived artefacts such as search indexes, cached exports and clips
NFR-12	Cross-border storage and processing are disclosed in the consent wording, since all data is hosted in the EU and GPU work may happen elsewhere CHANGED
NFR-13	Retention is enforced by Object Storage lifecycle rules plus a daily cleanup job: raw footage and HLS at 7 days, un-favourited clips at 14 days CHANGED
NFR-14	The Wideline company registers an information officer, and publishes a PAIA manual and privacy notice before the first pilot club joins. This and NFR-15 are gates on the v0 release, not engineering work, and need an owner outside the build team CHANGED
NFR-15	Every club signs Wideline's club terms, and every operator (Hetzner, GPU provider, email provider) signs an operator agreement. Every scout accepts the reviewer confidentiality terms CHANGED

These are product requirements, not legal advice. The consent wording, the reviewer terms, the responsible-party position and the cross-border position should be confirmed with a South African privacy lawyer before the first pilot club joins.

Security

ID	Requirement
NFR-16	Every query is scoped by organisation, or by review claim for scouts; cross-tenant access is impossible by construction, not by convention
NFR-17	Video is private; playback uses short-lived signed URLs
NFR-18	Data is encrypted at rest and in transit
NFR-19	Secrets are held in an encrypted secrets file or the CI secret store, injected at deploy; none in source control CHANGED
NFR-20	Role changes, exports, consent changes, review sessions and deletions are written to the audit log
NFR-21	Dependency and container scanning runs in CI; builds fail on high-severity findings
NFR-22	The server is firewalled to ports 22 (key-only, restricted IPs), 80 and 443; the database is never exposed publicly; OS security updates install automatically NEW

Accessibility and reach

ID	Requirement
NFR-23	WCAG 2.2 AA: keyboard operation, visible focus, 4.5:1 text contrast, labelled controls
NFR-24	The app works at 375 px, 768 px and 1280 px; review and search work on a phone
NFR-25	Video players have captions support and keyboard controls
NFR-26	Uploads tolerate poor connectivity: resumable, with clear progress and retry
NFR-27	No colour-only meaning, since kit colours already carry meaning in the UI

Reliability

ID	Requirement
NFR-28	99% monthly uptime for the web app during the pilot, accepting a single server; 99.5% from v1 CHANGED
NFR-29	Nightly database dumps to Object Storage plus Hetzner server backups, 30-day retention, with a restore tested quarterly CHANGED
NFR-30	Jobs that fail 3 times move to a dead-letter state and alert a platform admin
NFR-31	Structured logging, with a job's full history reconstructable
NFR-32	Raw footage is never deleted before 7 days have passed, and never deleted by any process other than the retention job CHANGED

Pricing and billing

Everything is free through v0 and v1. The point of the free period is to build the pool of consented players, because a scouting network with few players in it is worth nothing to a scout. Billing switches on at the v2 gate, once scouts have shown they will pay.

The engineering consequence matters more than the pricing: build the entitlement layer from day one, and switch on the payment provider later. Every limit the app enforces reads from an entitlement, never from a hard-coded constant or a role check. Retrofitting that into a live product is far more expensive than carrying it from the start.

Plan shape, to be priced later

Plan	Who	Switched on	Likely limits
Pilot	Invited academies	v0, free forever for pilot clubs	Generous; effectively unmetered
Club Free	Any academy or amateur club	v1, free	Matches per month, favourited clips
Club	Clubs wanting more volume	v2, paid	Higher match volume, several seats, more favourited clips
Scout	Individual scouts	v2, paid	Search, shortlists, reports, clip access. Active reviewers earn free months
Agency	Agents and professional clubs	v2, paid	Multiple seats, exports, saved-search alerts
Player	Players and guardians	Free permanently	Own profile, consent control, share link, favourite own clips

Players and guardians never pay. Charging the family of a child whose data the product depends on would be both bad business and a bad look.

Billing requirements

- Entitlements resolve as plan defaults, then per-organisation overrides, then feature flags. One function answers "may this organisation do this?"
- The payment provider sits behind an interface with one implementation per provider, so the choice can change without touching the product code.
- For a South African, ZAR-denominated product the realistic options are Paystack, Stripe, PayFast or Peach Payments. The decision should be made just before v2, not now, and should turn on recurring-billing support, ZAR settlement and fees at the volumes then expected.
- Usage is metered from v0 even though nothing is charged, so that pricing can be set from real cost data rather than guesswork.
- An organisation's data is never deleted because a subscription lapsed. It becomes read-only instead.

Risks and open questions

Risks

Risk	Impact	What reduces it
Player identity is unreliable: shirt numbers are rarely readable in wide-angle footage	High. Stats attached to the wrong player destroy trust	Lineup upload plus an explicit track-linking step confirmed by the reviewer, before any event is attributed
Not enough scouts review in time NEW	High. Unreviewed matches lose their raw footage after 7 days and never publish	Expiry-ordered queue and alerts (REVIEW-13); reviewer incentives; platform admins review as a fallback during the pilot
Reviewer bias or misconduct NEW	High. A scout could inflate or deflate a player's stats, or misuse footage	Conflict blocking (REVIEW-02), weekly spot-checks and club disputes (REVIEW-12), watermarked stream-only playback, suspension
Footage quality varies by club	High. A bad camera position breaks calibration and tracking	Minimum recording standards, validation at upload, and a per-venue calibration that is checked once and reused
Review burden stays high	High. If review takes 2 hours a match, scouts stop reviewing	Measure reviewed items per match from day one; lower the auto-accept thresholds only when measured recall justifies it
Model licensing	High. Shipping AGPL code in a SaaS forces open-sourcing or a licence fee	Apache-2.0 models only, with a licence review before any model enters the build
POPIA exposure around children's data	High. Regulatory and reputational, now carried directly by the Wideline company	Consent before processing, reviewer confidentiality terms, operator agreements, and a lawyer's review before the first pilot club joins
Single small server NEW	Medium. An outage or a full disk takes the whole product down	Nightly backups, Terraform-described setup so a rebuild takes under an hour, disk alerts, resize before v1
GPU cost per match	Medium. Could make the unit economics impossible	Hourly rental, automatic shutdown (NFR-07), cost metered per job, frames sampled rather than all processed
Cold start, two-sided	Medium. Scouts will not come for 5 clubs; clubs will not film for 2 scouts	Review work gives scouts early, exclusive sight of players, so they have a reason to join in v0
Ball tracking in wide-angle footage	Medium. The ball is only a few pixels wide	Accept it; use possession and player-proximity heuristics rather than relying on continuous ball tracking

Lost video evidence CHANGED	Medium. After 14 days most clips are gone, so old stats can't be re-checked or disputed	Encourage favouriting key clips; keep event timestamps and correction history; disputes must be raised within 7 days
Cross-border storage CHANGED	Medium. All data is held in the EU, not South Africa	Disclose it in consent wording, operator agreements with Hetzner and the GPU provider, lawyer's sign-off

Open questions

- Is "Wideline" clear of trademarks and available as a domain in South Africa? Alternatives if not: Unscouted, Touchline, Spotted.
- Which clubs are the pilot five, and what do they film with today? To be decided.
- What is the company's registered legal name, and who is the information officer?
- What do scouts get for reviewing? Options: early access to players, free months of the Scout plan from v2, or payment per match.
- Which hourly GPU provider? Criteria: price per hour for a T4-class or better GPU, start-up by API, billing by the minute, and region.
- Can a scout who reviewed a match contact a non-consented player's club about that player? This needs a clear rule in the reviewer terms.
- When a departed player joins a new club that also uses Wideline, who confirms the profile claim: the player or guardian, or the new club?

Decisions log

Topic	Decision (Oct 19, 2026)
Product name	Wideline, pending trademark and domain checks
Pilot clubs	To be decided
Ownership and POPIA	The Wideline company runs the product and is the responsible party. No other company is involved
Raw footage retention	Raw video deleted after 1 week. Clips kept for 2 weeks, then only favourited clips remain
Who reviews	Any registered scout, with verification, confidentiality terms and conflict rules
Hosting	A Hetzner server at the cheapest tier that fits, with GPUs rented by the hour per job. No Azure
Women's and girls' football	No different handling
Player leaves a club	Profile deactivated from the club; no data deleted

Sources

- Hetzner data centre locations (Germany, Finland, US, Singapore) and 2026 price changes: hosting reviews from Better Stack and Gappsy.
- Hetzner GPU range: dedicated GEX servers billed monthly; the GEX44 is no longer orderable (WZ-IT comparison, autumn 2026).
- Ultralytics licensing: AGPL-3.0 terms and when an Enterprise licence is required.
- RF-DETR: Apache-2.0 detection model, free to use commercially.
- POPIA sections 34 and 35: the prohibition on processing children's personal information, and its exceptions.

The cost, timing and accuracy targets in this document are estimates to be tested during the pilot, not sourced figures.